

Project 2.3.3 Robots for Good: Assistive Robotics

PLTW

ROBOTS FOR GOOD: ASSISTIVE ROBOTICS



"Painting" Coral Reefs?



Teachable Machine



Challenge Description



Research Scientists



Conclusion

PROJECT SUPPORTS



Source

“Painting” Coral Reefs?

GOALS	MATERIALS	RESOURCES
• Apply your knowledge of artificial intelligence and robotics to solve an authentic design problem.		
GOALS	MATERIALS	RESOURCES
• VEX® V5 POE Kit		
GOALS	MATERIALS	RESOURCES

- [Project 2.3.3 Robots for Good: Assistive Robots Rubric](#)
- [Teachable Machine](#)
- [VEX® Parts](#)
- [VEX® V5 Build Support Guides](#)
- [VEX® V5 Troubleshooting](#)
- [Python® Reference](#)
- [Project 2.3.3 Robots for Good: Assistive Robots \(Downloadable PDF\)](#)

Coral reefs are some of the most diverse ecosystems on the planet. They provide food and habitat for about 25% of the oceans' fish. Sadly, coral reefs are dying at unprecedented rates due in part to rising ocean temperatures. It is hard to know the full extent of the damage because scientists do not have an accurate estimate of how much healthy reef currently exists.

Researchers funded by the National Aeronautics and Space Administration (NASA) and others have developed a special camera that takes three-dimensional images of the ocean floor to identify coral. If the researchers were to look at all the images to identify which have coral and which do not, it would take them about two million years! However, the scientists developed a simple video game that invites people who care about coral reefs (or people that like games and just want to help!) to look at these images and “paint” over coral they see on the images to

train a machine-learning model to differentiate between coral and other underwater objects, like rocks.



Over time, the model will have had enough training to be able to map the images itself. When all of the world's coral has been mapped and the participatory science-trained machine learning model is fully functional, scientists will be better equipped to understand how climate change is affecting these unique ecosystems. View the ScienceCasts: Exploring Beneath the Waves video to learn more about this innovative [participatory science](#) project.

ScienceCasts: Exploring Beneath the Waves video¹

Scientists are just scratching the surface of possible applications of artificial intelligence (AI). In this project, you'll develop your own application using a powerful machine-learning tool called Teachable Machine.

Teachable Machine

Teachable Machine is a web-based program designed to help people develop and train machine-learning models for use in other software applications. To become familiar with the software, you'll make a simple model to detect something on the computer's camera. Here are some examples that you can try:

- Are you wearing a hat or not?
- Right side of your face, or left side of your face?
- You or your classmate?

Steps 1–10 walk through the example of holding a book in front of the camera or not, but you can try any of the examples, or make up your own classes.

1

Navigate to [Teachable Machine](#).

2

In the Teachable Machine interface, click the pencil icon for **Class 1** and type “book”.

3

Select the **Webcam** button. Make sure you have a neutral background, are facing your camera, and have a book in your hand in front of the camera. If prompted, agree to let the computer use your camera, or select the correct camera.

4

Press and hold the **Hold to Record** button until you've recorded at least 50 image samples; slightly vary your pose and the position of the book while recording.

NOTE: Place the book out of the camera's view when finished.

5

For **Class 2**, click the pencil icon and type "no book".

6

Select the **Webcam** button.

7

Press and hold the **Hold to Record** button. Be sure to record at least 50 image samples, slightly varying your pose and position within the frame.

8

When the model finishes training, it runs. To train your model, select the **Train Model** button.

9

To test your model, hold a book in the frame or not, and observe the Output of the model in the Preview box.

10

If the model does not work very well, you can continue training it by recording more image samples.



IMPORTANT: In steps 1–10, you used the Image Project in Teachable Machine. Two other projects are available: an Audio Project, which analyzes sound, and a Pose Project, which analyzes body positions.

Challenge Description

In this project, you'll develop an assistive robot using the VEX® V5 POE Kit that also uses machine learning. This could be a tool to assist a differently-abled family member or friend, a device that keeps a baby safe from some sort of hazard, or something else entirely. Whatever the context, you will first train a machine-learning model to differentiate between two conditions, assigning the conditions a value of green or red.

Then you reference that machine-learning algorithm in another program provided to you in p5.js, a free and open-source program for creative projects. This program displays a green or red box on your screen depending upon what your machine-learning algorithm concludes. The VEX V5 Optical Sensor will detect the green or red box and have your VEX robot react accordingly. Steps 11–16 walk you through the process of creating an algorithm and referencing the algorithm in p5.js.

11

With your team, define a problem and brainstorm solutions for what your assistive robot will do. Your robot must demonstrate two different behaviors depending upon input from an optical sensor. Seek guidance and

approval from your instructor before you move to step 12.

12

Train Teachable Machine to differentiate between your two conditions.

- Change the **Class 1** label to “Green”.
- Change the **Class 2** label to “Red”.
- Record your images for each of the two classes.
- Train your model.
- Preview your model to ensure it functions as you intend. Modify and continue training if necessary.

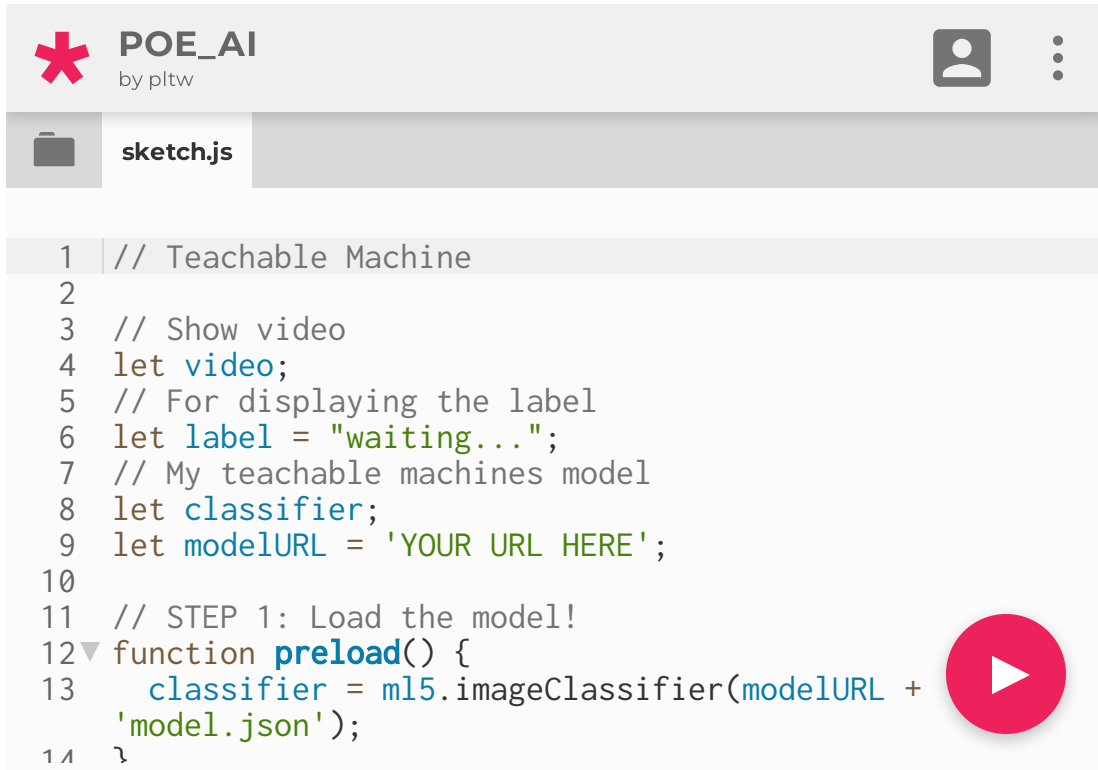
13

Export your model.

- Select **Export Model** next to Preview.
- In the box that opens, make sure **Upload (shareable link)** is selected.
- Select **Upload my model**.
- Select **Copy**.

14

Study the following p5.js Editor interactive. Select **Allow All** cookies.



```
1 // Teachable Machine
2
3 // Show video
4 let video;
5 // For displaying the label
6 let label = "waiting...";
7 // My teachable machines model
8 let classifier;
9 let modelURL = 'YOUR URL HERE';
10
11 // STEP 1: Load the model!
12 function preload() {
13   classifier = ml5.imageClassifier(modelURL +
14   'model.json');
15 }
```

Open the [p5.js model](#) in a new browser tab.

15

In line 9 of the code in the p5.js interactive, replace the text '*YOUR URL HERE*' with the shareable link you copied in step 13. Be sure to keep the single quotes before and after the link.

16

Test the integration of your Teachable Machine model in p5.js.

- In the top left of the p5.js window, select the **Play** icon. Test your machine learning model. For example, if it is listening for a sound, make the sound.

- Confirm that p5.js outputs either a green or red box depending upon the input.

17

Design and construct your robot to include the Optical Sensor. Position the sensor in front of your screen to obtain readings from the p5.js window. Program the sensor so that your robot reacts differently depending on whether it senses green or red.

NOTE: If you need help during this step, review [VEX®](#) [Parts](#), [VEX® V5 Build Support Guides](#), [VEX® V5 Troubleshooting](#), and [Python® Reference](#).

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Share your robot with other teams. Provide an explanation of your intended client and how your design solves their problem.

Research Scientists



CAREER CONNECTION

Dr. Di Yang | *Research Scientist*

Originally from northeastern China, Di came to the United States for her graduate studies. As an international woman scholar in STEM, Di contributes to intercultural education and a deeper appreciation of diversity and global issues. Read more about how she combines citizen science data and artificial intelligence in her research.

I am currently an Assistant Professor at University of Wyoming's Geographic Information Science Center (WyGISC). My research is focused on geospatial informatics—the science and technology of gathering, storing, analyzing, interpreting, modeling, distributing, and using spatially referenced information—and applied geospatial artificial intelligence, or GeoAI. It's a mouthful, I know! Let me break it down.



I use Earth observation data collected by citizen scientists to train machine learning tools. Humans are amazing at analyzing images, but it would be impossible to analyze all those satellite images of Earth. Instead, if we use humans to train computers with a subset of satellite images, and then let the computers analyze the millions of images, we have a better shot of fully analyzing all those images.

The resulting data analysis can help guide land management decisions (think water resources, or areas of fire vulnerability). Ultimately, I hope to help humanity gain a quantitative understanding of the relationships between humans, landscape patterns, and ecological processes.

I have a deep appreciation for the natural world as well as other cultures. Outside of my career, I am passionate about Chinese ink painting, which is done by brushing black ink or colored pigments onto paper or silk.

STEM research has taken me all over the world, and has made for some memorable stories. Perhaps most notably, I was attacked by a bison while conducting research in Florida as part of my PhD program! Word to the wise: Bison are not the most friendly animals! What I like most about my STEM career is that I am constantly immersed in different natural environments, and I get to use my training and insights to develop tools to protect them.



Parting advice from Dr. Yang:

- Write down all of your brilliant ideas while you're still in school, because that's your most creative age, and you might get the chance to build on them later.

- Celebrate every milestone you meet. It can help keep you motivated and remind you of the progress you've made so far, and it's important to take breaks!



Reflection



Visit [Zooniverse](#) (a large collection of people-powered research projects) and scroll through some of the projects.

- Which projects interest you most?
- Which projects surprise you?
- What are some of the pros and cons to using people-powered research?
- Do the pros outweigh the cons?

Conclusion

Question 1

Which team had the most innovative application of AI? Why?

Source

Third-party images, videos, and Historical Highlight information used throughout this project were drawn from the following source:

1. Exploring Beneath the Waves, NASA ScienceCasts, NASA, permitted under [**CC BY-SA 3.0**](#)